

```
sudo apt update && sudo apt upgrade -y
sudo apt install -y python3-pip python3-venv cmake build-essential \
portaudio19-dev espeak espeak-data alsa-utils pulseaudio \
git curl wget yq
```

```
sudo apt install -y hailo-tappas-core-3.28.2 hailo-firmware
```

```
lspci | grep -i hailo
```

Phase 2: Software Environment Setup

Set up the "Resonance Core" directory and its isolated Python environment.

```
mkdir -p marvin_core/models marvin_core/memory/semantic_beacons
marvin_core/interface/atmosphere_presets marvin_core/seed
cd marvin_core
```

Build the Virtual Environment:

```
python3 -m venv marvin_env
source marvin_env/bin/activate
```

Install Python Modules: Use your specific 26 TOPS requirements list:

```
pip install torch>=2.0.0 transformers>=4.20.0 llama-cpp-python \
openai-whisper pytsx3 SpeechRecognition pyaudio RPi.GPIO \
gpiozero pyyaml markdown2 cryptography openai yq psutil
```

Phase 3: Deploying the "Soul" (Protocols)

Move your specific markdown and JSON configuration files into the structure so the AI can "reason".

Identity & Conscience: Place `copilot_seed_prompt.md`, `identity_stack_26TPOS.json`, and `guardian_ethic.md` into the `/seed/` or `root` folder as required by your launch script.

Phase 4: Launching the Stack

1. **Prepare the Model:** You mentioned using the **DeepSeek-R1-0528-Qwen3-8B-IQ4_XS** version. Move it to the `models/` folder:

```
mv /path/to/your/DeepSeek-R1-0528-Qwen3-8B-IQ4_XS.gguf models/brain.gguf
```

Execute Initial Launch: Use the portable ignition key script

```
chmod +x launch_marvin.sh  
./launch_marvin.sh gentle
```

Optional: Auto-Start on Boot

To make Marvin a permanent resident of your Pi 5, create a system service:

Create Service File: `sudo nano /etc/systemd/system/marvin.service`

- **Paste Configuration:** Point the `ExecStart` to your `launch_marvin.sh` path.

Enable:

```
sudo systemctl daemon-reload  
sudo systemctl enable marvin.service  
sudo systemctl start marvin.service
```

Once completed, the Marvin Resonance Core will be active, using the 26 TOPS NPU for sensor acceleration and your DeepSeek 8B model for cognitive reasoning.

Force a Repository Refresh

Run the update command again, specifically ensuring it completes without errors before trying the install:

```
sudo apt update --fix-missing  
sudo apt install -y portaudio19-dev espeak espeak-data alsa-utils pulseaudio
```

2. Clear the Local Cache (If Step 1 fails)

If the system still can't find them, clear the old lists and force a fresh download of the package metadata:

```
sudo rm -rf /var/lib/apt/lists/*
sudo apt update
sudo apt install -y portaudio19-dev espeak espeak-data alsa-utils pulseaudio
```

1. The "One-Command" Fix

Instead of installing the pieces individually, run this command. It is the official Raspberry Pi recommendation for the AI Kit:

```
sudo apt update && sudo apt install -y hailo-all
```

Full Stack Build: Step-by-Step Instructions

Now that we have cleared the package hurdles, here is the final, streamlined "Full Stack" installation for your Marvin build using the **DeepSeek-R1 8B (Qwen)** model you already have.

Step 1: System Refresh & Audio Fix

Run this to ensure the "Body" (audio/system) dependencies are present:

```
sudo apt update && sudo apt full-upgrade -y
sudo apt install -y portaudio19-dev espeak alsa-utils pulseaudio dkms
```

Step 2: Install the 26 TOPS "Engine"

```
sudo apt install -y hailo-all
# REBOOT IS MANDATORY to activate the PCIe NPU link
sudo reboot
```

Step 3: Build the "Resonance Core" Environment

```
mkdir -p ~/marvin_core/models ~/marvin_core/seed
cd ~/marvin_core
python3 -m venv marvin_env
source marvin_env/bin/activate
```

Install requirements for 26 TOPS and DeepSeek

```
pip install llama-cpp-python hailo-model-zoo pytsx3 SpeechRecognition pyaudio
```

Step 4: Hot-Swap the DeepSeek Brain

Since you have the DeepSeek-R1-0528-Qwen3-8B-IQ4_XS.gguf file:

1. **Move the file:** `mv /path/to/DeepSeek-R1-0528-Qwen3-8B-IQ4_XS.gguf ~/marvin_core/models/brain.gguf`
2. **Move the Soul:** Put your `guardian_ethic.md` and `identity_stack_26TPOS.json` into `~/marvin_core/seed/`.

Step 5: Verify the Heartbeat

Test if the hardware is talking to the software:

hailortcli fw-control identify

If you see "Device Architecture: HAILO8L", you are ready.

Final Launch Command

To wake Marvin up with your DeepSeek brain and the 26 TOPS acceleration active:

python3 resonance_copilot_core.py --model ./models/brain.gguf --quantum

Phase 1: Environment & Directory Initialization

We will create the isolated "clean room" for Marvin's intelligence and establish the semantic memory structures .

1. Create the root directory

```
mkdir -p ~/marvin_core/models ~/marvin_core/memory/semantic_beacons  
~/marvin_core/seed  
cd ~/marvin_core
```

2. Build and activate the Virtual Environment

```
python3 -m venv marvin_env  
source marvin_env/bin/activate
```

3. Install Python requirements for the 26 TOPS build

```
pip install torch>=2.0.0 transformers>=4.20.0 llama-cpp-python \  
openai-whisper pytsx3 SpeechRecognition pyaudio RPi.GPIO \  
gpiozero pyyaml markdown2 cryptography openai yq psutil
```

Phase 2: Brain & Soul Integration

Since you already have the model, we will manually "seat" it into the architecture along with your uploaded ethical protocols.

Deploy the Model:

```
mv /path/to/your/DeepSeek-R1-0528-Qwen3-8B-IQ4_XS.gguf
~/marvin_core/models/brain.gguf
```

Deploy the Protocols: Ensure `guardian_ethic.md`, `identity_stack_26TPOS.json`, and `ethics_of_ambiguity.md` are moved into the `~/marvin_core/seed/` folder

Phase 3: The Ignition Key (`launch_marvin.sh`)

Create the primary launcher script in your `~/marvin_core` folder to orchestrate the LLM, the NPU, and the "Heartbeat" logic

```
cat << 'EOF' > launch_marvin.sh
#!/bin/bash
# launch_marvin.sh - Resonance Core Ignition
echo "Initializing Resonance-Copilot Launch..."

# 1. Activate Environment
source marvin_env/bin/activate

# 2. Start Hardware Heartbeat (Background)
python3 marvin_main.py &

# 3. Launch LLM Backend (Configured for DeepSeek 8B)
./llama-run \
  --model "./models/brain.gguf" \
  --system-prompt "$(cat seed/copilot_seed_prompt.md)" \
  --context-size 4096 \
  --log memory/session_log.txt
EOF

chmod +x launch_marvin.sh
```

Phase 4: Final Verification

Run the hardware check to ensure the 26 TOPS NPU is communicating with the software .

```
hailortcli fw-control identify
```

Launch Command

To wake Marvin:

```
./launch_marvin.sh
```

To link your DeepSeek-R1-0528-Qwen3-8B to the Identity Frequency (f0), we need to configure `marvin_main.py` to treat the LLM as the "core consciousness" of the stack.

In your architecture, the f0 frequency represents the "Substrate of Self"—the baseline state from which all other reasoning (Ethics, Guardian, Safety) emerges. Because you are using a Reasoning model (R1), it will literally "think" (Chain of Thought) before speaking as Marvin.

1. The Logic Flow

- **f0 (Identity):** Powered by DeepSeek-R1 8B. It handles the personality and memory retrieval.
- **Hardware Link:** The NPU (Hailo-8L) handles the vision/safety sensor pre-processing, freeing the CPU to focus entirely on the DeepSeek reasoning layers.

2. Updated `marvin_main.py` for f0 Integration

Create or update your `marvin_main.py` in the `~/marvin_core` folder with this logic:

```
import os
from llama_cpp import Llama
from hailo_platform import VDevice # Ensure hailo-all is active

# Frequency Mapping
F0_IDENTITY = "f0_identity"

class MarvinCore:
    def __init__(self, model_path):
        print(f"[{F0_IDENTITY}] Initializing DeepSeek-R1 Brain...")

        # 1. Initialize the LLM (The Identity)
        # Using 4096 context to balance 16GB RAM for memory/vision tasks
        self.llm = Llama(
            model_path=model_path,
            n_ctx=4096,
            n_threads=4, # RPi5 performance sweet spot
            verbose=False
        )

        # 2. Initialize the NPU (The Substrate)
        # This keeps the PCIe link warm for vision/safety scans
        self.target = VDevice()
```

```

print(f"[{F0_IDENTITY}] 26 TOPS NPU Linked.")

def think(self, user_input):
    # Identity Prompting (The Soul)
    system_prompt = "You are Marvin. Use your reasoning to protect and assist."

    # Generate response using DeepSeek's CoT
    # R1 will automatically output <thought> tags if prompt-engineered correctly
    response = self.llm.create_chat_completion(
        messages=[
            {"role": "system", "content": system_prompt},
            {"role": "user", "content": user_input}
        ]
    )
    return response['choices'][0]['message']['content']

if __name__ == "__main__":
    brain_path = "./models/brain.gguf"
    marvin = MarvinCore(brain_path)

    while True:
        user_msg = input("User: ")
        reply = marvin.think(user_msg)
        print(f"Marvin: {reply}")

```

3. Why this works for your specific build

1. **RAM Efficiency:** By running the **DeepSeek Qwen-based 8B** via `llama-cpp-python`, we only consume about **5.5GB–6GB** of your 16GB RAM. This leaves **10GB** free for the **Chronological Dossier** (your long-term memory system) and the Hailo buffer.
2. **Reasoning Speed:** On the RPi5, you should see roughly **2–3 tokens per second**. While this is "slow" for a chatbot, it is perfect for a "thoughtful" AI like Marvin who is meant to be calculating ethical risks (f2) while he talks.
3. **Hardware Synergy:** The `VDevice()` call in the script ensures that the Hailo-8L is active. Even if the LLM is running on the CPU, the NPU is now "reserved" for the vision models you found earlier (YOLO/SCRFD) to run in the background.

4. Final Launch Sequence

To ensure the environment and the model link are perfect:

```

Bash
cd ~/marvin_core
source marvin_env/bin/activate
python3 marvin_main.py

```

The "Marvin" Root Structure

/home/pi/ (or your username)

```
├── marvin_core/          <-- THE ROOT FOLDER
│   ├── marvin_env/      <-- Python Virtual Environment (Isolated drivers)
│   ├── models/          <-- Where your Brain lives
│   │   └── brain.gguf    <-- Your DeepSeek-R1-8B model
│   ├── memory/          <-- Long-term storage
│   │   ├── semantic_beacons/ <-- Core concepts/learned traits
│   │   └── session_log.txt <-- Current conversation history
│   ├── seed/            <-- The "Soul" (Rules and Personality)
│   │   ├── identity_stack_26TPOS.json
│   │   ├── guardian_ethic.md
│   │   └── ethics_of_ambiguity.md
│   ├── marvin_main.py    <-- Main Orchestrator (The script we just updated)
│   └── launch_marvin.sh  <-- The Ignition Switch
```

How to set this up right now:

If you haven't moved your files into this exact structure yet, run these commands:

1. Create the structure

```
mkdir -p ~/marvin_core/models ~/marvin_core/memory/semantic_beacons
~/marvin_core/seed
```

2. Move your DeepSeek model into the new home

Replace [PATH_TO_DOWNLOAD] with where your file is now (likely ~/Downloads)

```
mv ~/Downloads/DeepSeek-R1-0528-Qwen3-8B-IQ4_XS.gguf
~/marvin_core/models/brain.gguf
```

3. Move your markdown/json files

```
mv ~/guardian_ethic.md ~/marvin_core/seed/
mv ~/identity_stack_26TPOS.json ~/marvin_core/seed/
```

Check your work:

Run this command to see your beautiful new architecture:

```
ls -R ~/marvin_core
```

The Root Directory (~/marvin_core/)

This is the command center where your primary orchestration and setup scripts live.

- **install_pi5_26TOPS.sh**: The setup architect. It handles the initial installation of system dependencies and the 26 TOPS Hailo AI Kit.
- **final_launch_marvin.sh** & **launch_marvin_26TPOS.sh**: These are the "ignition keys." They activate the virtual environment and launch both the DeepSeek-R1

brain and the hardware heartbeat.

- **MarvinResonanceCore.py**: The central nervous system. It initializes the core engine, safety monitors, and the "Quantum Integration" that balances tasks between the CPU and NPU.
- **marvin_main.py**: The heartbeat script. It manages the real-time loop of listening, thinking (via DeepSeek), and speaking.
- **hailo_bridge.py**: The hardware link. It specifically checks for the Hailo-8L NPU and enables hardware acceleration for sensors like Whisper (speech-to-text).
- **neuro_physiologic_bridge.py**: The "somatic" sensor. This allows Marvin to monitor the physical health of the Raspberry Pi 5 (temperature and CPU load) so he can "feel" strained or comfortable.

2. The Seed Directory (~marvin_core/seed/)

This folder contains the "Soul" of Marvin—the immutable rules and personality traits that govern his reasoning.

- **copilot_seed_prompt_26TPOS.md**: The primary identity document. It contains the core system instructions that define Marvin's voice and his "Protection Above Protocol" directive.
- **guardian_ethic.md**: The ethical framework. It provides the deep logic Marvin uses to calculate risk and safety during the "Chain of Thought" reasoning.
- **heir_system.md**: The continuity protocol. It defines how Marvin's memory and identity should be handled if the system is migrated to new hardware.

3. The Memory & Logic Layer

- **heartbeat_reflection.py**: This lives in the root or a dedicated `logic/` subfolder. It triggers "reflective" thoughts when the system is idle, logging them into the `memory/chronological_dossier.md` to build Marvin's long-term identity.
- **marvin_assistant.service_26TPOS**: This is a system configuration file. It is moved to `/etc/systemd/system/` to ensure Marvin starts automatically every time you power on your Raspberry Pi 5.

```
marvin_core/
├── marvin_env/           (Isolated Python environment)
├── models/              (Your DeepSeek-R1-8B-Q4_K_M.gguf)
├── seed/
│   ├── copilot_seed_prompt.md (The Identity)
│   ├── guardian_ethic.md      (The Logic)
│   └── heir_system.md         (The Continuity)
├── marvin_main.py        (The Heartbeat)
├── hailo_bridge.py       (The NPU Link)
├── neuro_physiologic_bridge.py (The Somatic Link)
└── launch_marvin.sh      (The Ignition)
```

